Physics - Graduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. According to Hooke's law, if you hang by a tree branch and note how much it bends, then hanging with twice the weight produces
   (a) three times the bend.
   (b) twice the bend.
   (c) four times the bend if the branch doesn't break.
   (d) no bend at all.
   (e) a bend that is unpredictable.

2. Thermal conduction involves mainly
   (a) isotopes
   (b) ions
   (c) mesons
   (d) protons
   (e) photons

3. Aplano-convex lens in air has a power of 5 diopters. What is the power of the lens in water? The index of refraction of the glass in the lens is 1.60.
   (a) 0.83 diopters
   (b) 3.3 diopters
   (c) 5.0 diopters
   (d) 1.0 diopters
   (e) 1.2 diopters

4. How long would it take a car, starting from rest and accelerating uniformly in a straight line at 5 m/s^2, to cover a distance of 200 m ?
   (a) 18.0 s
   (b) 15.5 s
   (c) 9.0 s
   (d) 8.0 s
   (e) 10.5 s

5. Anucleusof mass M emits a Υ ray of energy E_$\Upsilon^{217}. Thenucleuswasinitiallyatrest. Whatisitsrecoilen$

6. E_^{2}/4Mc^2

7. $2E_\Upsilon^2/Mc^2$
8. E_Υ/Mc^2
9. $3E_\Upsilon^2/2Mc^2$
10. $E_\Upsilon/4Mc^2$

True / False (10 marks)

Answer True or False

6. True or False: The force exerted on the block by the table is parallel to the surface of the table during movement.
7. True or False: The atomic number of the resulting element changes by minus 3 after the emission of 1 alpha particle and 2 beta particles.
8. True or False: The object distance for a concave mirror producing an image at -16.2 m with a magnification of 1.79 is 11.2 m.
9. True or False: The specific heat capacity of sand is higher than that of vegetation, leading to a slower cooling rate at night.
10. True or False: A common nuclear fission reactor heats water to produce steam for generating electricity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain the process of light intensity reduction as unpolarized light passes through two ideal linear polarizers oriented at a 45-degree angle to each other. What is the final intensity as a percentage of the incident intensity, and how do you derive this result?
12. Discuss the principles of projectile motion to determine the maximum height a body will reach when projected upward with an initial velocity of 1000 cm/s.

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